



CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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0625/61

May/June 2020

1 hour

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

This document has **12** pages. Blank pages are indicated.

- 1 A student investigates the balancing of a metre rule.

Fig. 1.1 shows the arrangement.

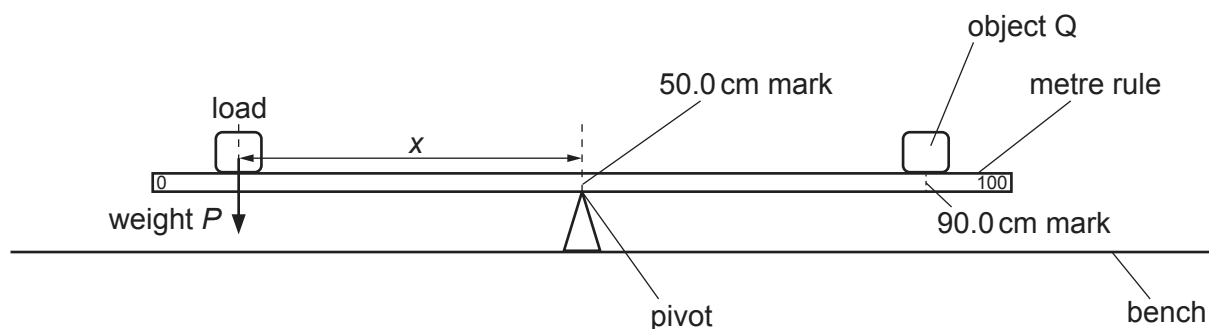


Fig. 1.1

- (a) The student places the metre rule on the pivot at the 50.0 cm mark. He places an object Q on the metre rule with its centre at the 90.0 cm mark. He places a load of weight $P = 2.0\text{ N}$ on the metre rule and adjusts the position of the load so that the metre rule is as near as possible to being balanced.

He measures the distance x from the centre of the load to the pivot.

He repeats the procedure using loads of weight $P = 3.0\text{ N}$, 4.0 N , 5.0 N and 6.0 N . All the values of P and x are recorded in Table 1.1.

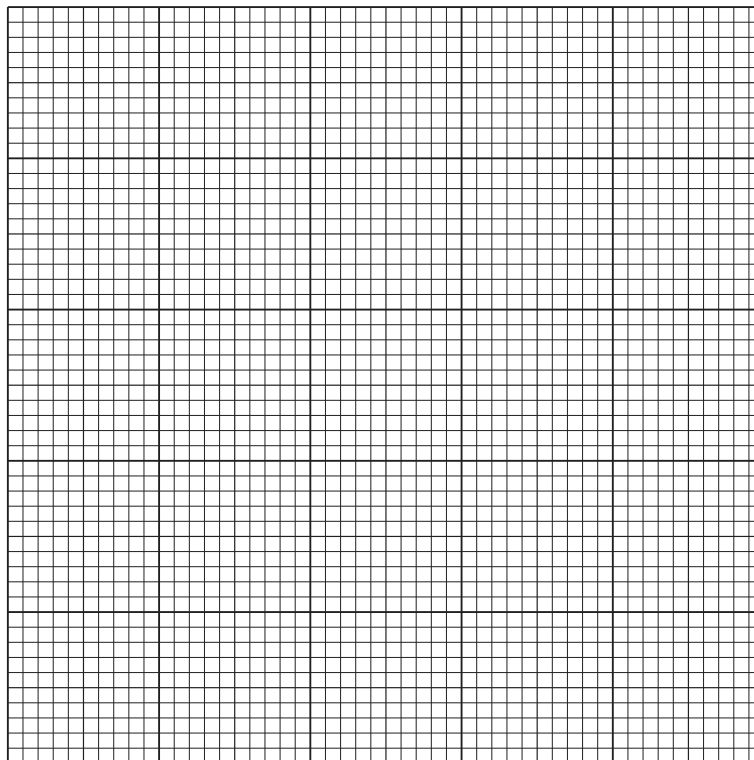
Table 1.1

P/N	x/cm	$\frac{1}{x} / \frac{1}{\text{cm}}$
2.0	40.0	
3.0	27.0	
4.0	20.0	
5.0	15.9	
6.0	13.3	

Calculate, and record in Table 1.1, the values of $\frac{1}{x}$.

[2]

- (b) Plot a graph of P/N (y -axis) against $\frac{1}{x} / \frac{1}{\text{cm}}$ (x -axis). Start both axes at the origin (0,0).



[4]

- (c) In this experiment, x_{max} , the maximum possible value for x is 50.0 cm. Calculate $\frac{1}{x_{\text{max}}}$.

$$\frac{1}{x_{\text{max}}} = \dots\dots\dots \frac{1}{\text{cm}}$$

Use the graph to determine the minimum value of P required to balance the metre rule in this experiment. Show clearly on the graph how you determined this value.

minimum value of $P = \dots\dots\dots$ [2]

- (d) In this experiment, the width of object Q is slightly greater than the width of the metre rule. Explain briefly how you would place the object Q as accurately as possible on the 90.0 cm mark of the metre rule. You may draw a diagram.

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..... [1]

- (e) In this experiment, it is difficult to determine the exact position of the load that will make the metre rule balance.

- (i) Explain briefly why this is difficult.

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..... [1]

- (ii) Explain briefly how you would find the best position of the load that will make the metre rule balance.

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..... [1]

[Total: 11]

- 2 A student determines the resistances of some filament lamps.

Fig. 2.1 shows the first circuit she uses.

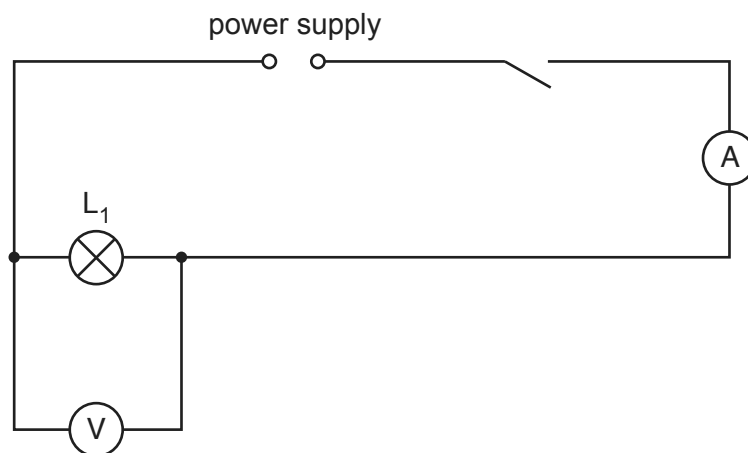


Fig. 2.1

- (a) (i) Record the potential difference V_1 across the lamp L_1 , as shown on the voltmeter in Fig. 2.2.

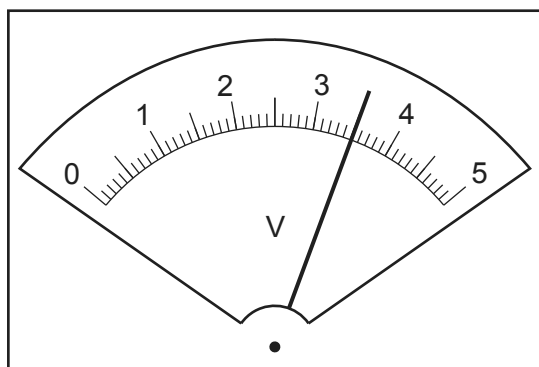


Fig. 2.2

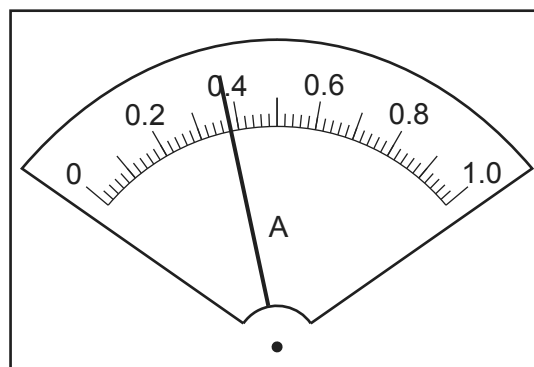


Fig. 2.3

$$V_1 = \dots\dots\dots \text{ V [1]}$$

- (ii) Record the current I_1 in the circuit, as shown in Fig. 2.3.

$$I_1 = \dots\dots\dots \text{ A [1]}$$

- (iii) Calculate the resistance R_1 of the filament of lamp L_1 . Use the equation $R_1 = \frac{V_1}{I_1}$. Include the unit.

$$R_1 = \dots\dots\dots [2]$$

- (b) The student disconnects the voltmeter. She connects lamp L_2 in series with lamp L_1 . She connects the voltmeter across lamp L_2 .

She measures the current I_2 in the circuit and the potential difference V_2 across lamp L_2 .

$$I_2 = \frac{0.30 \text{ A}}{\dots\dots\dots}$$

$$V_2 = \frac{1.7 \text{ V}}{\dots\dots\dots}$$

Calculate the resistance R_2 of the filament of lamp L_2 . Use the equation $R_2 = \frac{V_2}{I_2}$.

$$R_2 = \dots\dots\dots$$

The student disconnects the voltmeter. She connects lamp L_3 in series with lamps L_1 and L_2 . She connects the voltmeter across lamp L_3 .

She measures the current I_3 in the circuit and the potential difference V_3 across lamp L_3 .

$$I_3 = \frac{0.26 \text{ A}}{\dots\dots\dots}$$

$$V_3 = \frac{1.2 \text{ V}}{\dots\dots\dots}$$

Calculate the resistance R_3 of the filament of lamp L_3 . Use the equation $R_3 = \frac{V_3}{I_3}$.

$$R_3 = \dots\dots\dots [1]$$

- (c) Calculate $R_1 + R_2 + R_3$. Give your answer to a suitable number of significant figures for this experiment.

$$R_1 + R_2 + R_3 = \dots\dots\dots [1]$$

- (d) Some students make suggestions about the results of the experiment.

Suggestion **A**: $R_1 + R_2 + R_3$ should be equal to $3 \times R_1$.

Suggestion **B**: $R_1 + R_2 + R_3$ should be less than $3 \times R_1$.

Suggestion **C**: $R_1 + R_2 + R_3$ should be greater than $3 \times R_1$.

State which suggestion **A**, **B** or **C** agrees with your results. Justify your answer by reference to your results.

statement

justification

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[2]

- (e) Draw a circuit diagram to show the circuit used in part (b) with all three lamps connected in series.

[3]

[Total: 11]

- 3 A student investigates the position of the image in a plane mirror.

Fig. 3.1 shows the ray-trace sheet he uses.

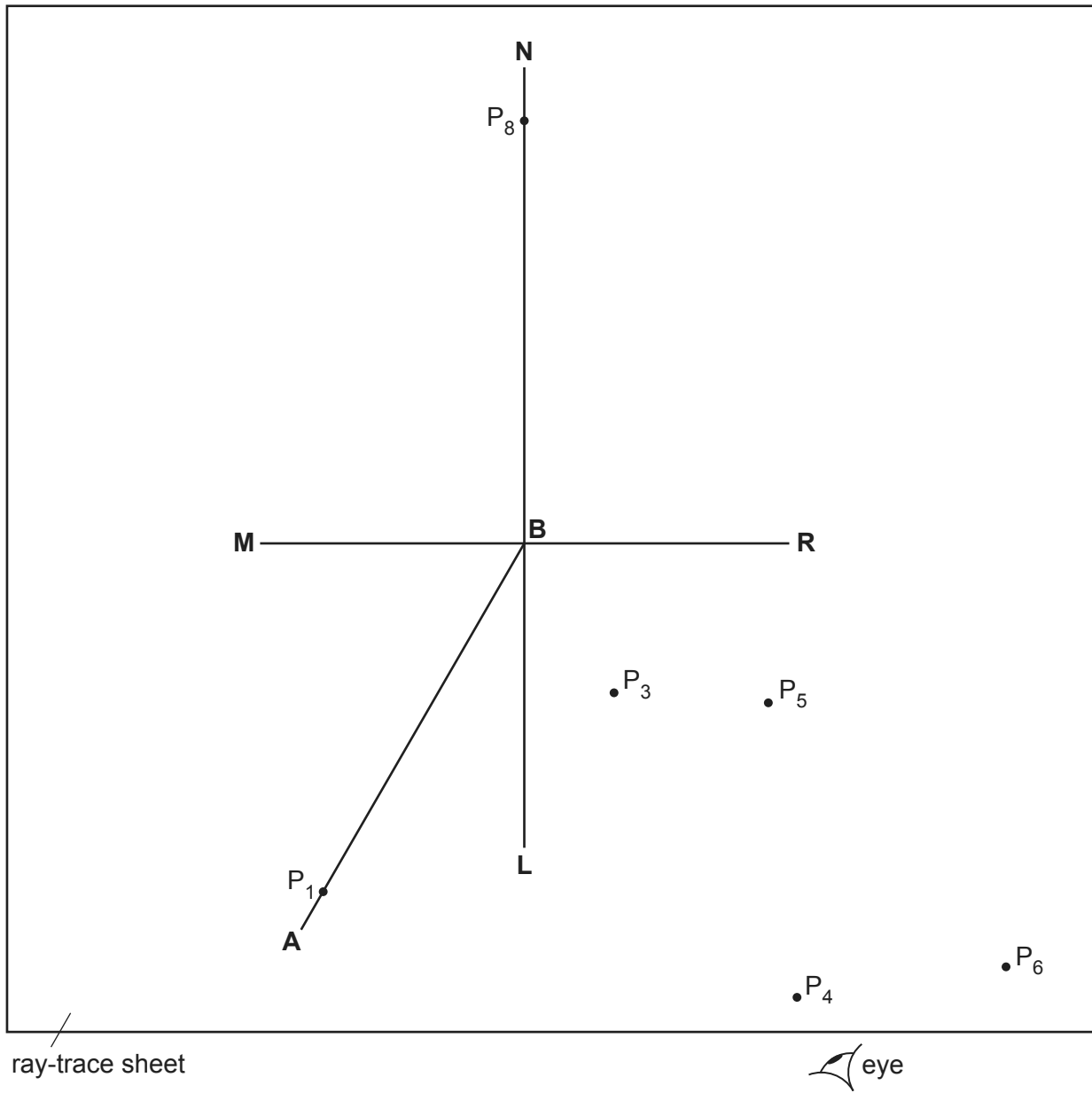


Fig. 3.1

(a)

- The student draws the line **MR**.
- He draws a normal **NL** to this line that passes through the centre of **MR**.
- He labels the point at which **NL** crosses **MR** with the letter **B**.
- He draws a line from **B** at an angle of incidence $i = 30^\circ$ to the normal below **MR** and to the left of the normal. He labels the end of this line **A**.
- He places a pin P_1 on line **AB**, as shown in Fig. 3.1. He places another pin P_2 on the line **AB**.
- He places the reflecting face of the mirror vertically on the line **MR**.
- He views the images of pins P_1 and P_2 from the direction indicated by the eye in Fig. 3.1.

(i) On Fig. 3.1, mark with a cross a suitable position for pin P_2 in this experiment. [1]

- He places two pins P_3 and P_4 some distance apart so that pin P_3 and the images of P_2 and P_1 all appear exactly behind pin P_4 . The positions of P_3 and P_4 are shown on Fig. 3.1.

(ii) Draw the line joining the positions of P_3 and P_4 . Continue the line until it extends at least 7.0 cm beyond **MR**. [2]

(b)

- The student keeps pin P_1 in the same position but moves pin P_2 so that the angle of incidence $i = 40^\circ$.
- The pin positions P_5 and P_6 for the reflected ray are marked on Fig. 3.1.

(i) Draw the line joining the positions of P_5 and P_6 . Continue the line until it extends at least 7.0 cm beyond **MR**.

Label with the letter **Y** the point where the two lines cross beyond **MR**. [1]

(ii) Draw a line from P_1 to **MR** that meets **MR** at a right angle. Measure and record the length a of this line.

$a = \dots\dots\dots$ [1]

(iii) Draw a line from the point labelled **Y** to **MR** that meets **MR** at a right angle. Measure and record the length b of this line.

$b = \dots\dots\dots$ [1]

(c)

- The student removes all the pins. He places pin P_7 on the normal at a distance 6.0 cm from the front of the mirror.
 - He views the image of P_7 in the mirror.
 - He places pin P_8 on the normal behind the mirror.
 - He adjusts the position of P_8 so that the image of the bottom of the pin P_7 and the top of pin P_8 seen over the mirror appear as one pin when viewed from all angles in front of the mirror.
- (i) On Fig. 3.1, measure the distance x along the normal between P_8 and the mirror.

$x = \dots\dots\dots$ [1]

- (ii) Complete the diagram in Fig. 3.2 to show the appearance of the image of pin P_7 and pin P_8 as described in (c).



Fig. 3.2

[1]

- (d) The student expects the readings to show that the image formed in a plane mirror is the same distance behind the mirror as the object is in front of the mirror. Readings of $a = b$ and $x = 6.0$ cm will show this.

State whether your readings show that the image formed in a plane mirror is the same distance behind the mirror as the object is in front of the mirror. Justify your statement by reference to the readings.

statement

justification

.....

[2]

- (e) The student carries out this experiment with care. Suggest a practical reason why the results may not be accurate.

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..... [1]

[Total: 11]

- 4 A student investigates the effect of the colour of the surface of a metal container on the rate of loss of heat from the container. She knows that black surfaces are better radiators of thermal energy than white surfaces and wants to investigate the effect of other colours.

The following apparatus is available:

metal containers each with the outer surface painted a different colour
a thermometer
a stop-watch
a supply of hot water.

She can also use other apparatus and materials that are usually available in a school laboratory.

Plan an experiment to investigate the effect of the colour of the surface of a metal container on the rate of loss of heat from the container.

You should:

- draw a diagram of the apparatus used
- explain briefly how you would carry out the investigation
- state the key variables to be kept constant
- draw a table, or tables, with column headings, to show how you would display your readings (you are **not** required to enter any readings in the table)
- explain how you would use your readings to reach a conclusion.

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